Module 12

Technical Workouts

1. Understand Unsupervised Learning Fundamentals

a. What is unsupervised learning
b. Learning from unlabeled data
c. Pattern discovery
d. Segmentation concepts
e. Real-world applications

2. Understand Clustering Algorithms

a. K-Means Clustering
b. Hierarchical Clustering
c. Cluster analysis
d. Customer segmentation
e. Cluster evaluation

3. Understand Dimensionality Reduction

a. High-dimensional data
b. PCA fundamentals
c. Feature compression
d. Visualization of high-dimensional data
e. Reducing complexity

4. Understand Feature Engineering

a. Feature creation
b. Feature transformation
c. Feature selection
d. Handling categorical data
e. Improving model quality

5. Understand Model Evaluation

a. Accuracy
b. Precision
c. Recall
d. F1 Score
e. Confusion Matrix

6. Understand Cross Validation

a. Why validation matters
b. K-Fold Cross Validation
c. Reliable model evaluation
d. Generalization assessment
e. Performance consistency

7. Understand Hyperparameter Tuning

a. Hyperparameters vs parameters
b. Grid Search
c. Random Search
d. Model optimization workflows
e. Improving performance

8. Understand End-to-End ML Pipelines

a. Data preparation
b. Feature engineering
c. Model training
d. Evaluation
e. Optimization